

Integer quadrature rounding gives $C(n!)^2$ fair dice

Research notes, 30 September 2026

Status. Complete proof with independent mathematical reviews completed by two collaborating agents. These reviews are not machine-checked formal verification. The asymptotic class remains $\exp(O(n \log n))$, but the leading coefficient in the exponent matches the whole-copy insertion barrier proved in the companion note.

Theorem 1. *There is an absolute positive integer C such that, for every $n \geq 1$, there are n equal-sided permutation-fair dice with*

$$F_n = C(n!)^2$$

faces each. One explicit admissible choice is $C = 2^{25}$. The construction adds one letter at a time, using $(r+1)^2$ unchanged copies in the step from r letters to $r+1$ letters.

The improvement comes from replacing common-denominator clearing of the specific Hahn weights by rounding and exact integer corrections in a unimodular binomial basis. The existing positivity estimates are retained.

Lemma 2 (Integer quadrature rounding). *Let $N \geq r \geq 1$, and suppose $w_0, \dots, w_N$ form an exact degree- r quadrature for the uniform probability measure on $[0, 1]$, supported at i/N , with $w_i \geq \eta > 0$. Define*

$$\mu_j = \int_0^1 \binom{Nx}{j} dx \quad (0 \leq j \leq r).$$

Suppose K is a positive integer such that all $K\mu_j$ are integers. If

$$K\eta \geq 1 + \max_{0 \leq i \leq r} \sum_{j=i}^r \binom{j}{i} \binom{N+1}{j+1},$$

then there are nonnegative integers $k_0, \dots, k_N$ with $\sum_i k_i = K$ whose normalized weights k_i/K give another exact degree- r quadrature on this grid.

Proof. Set $a_i = \lfloor Kw_i \rfloor$ and

$$b_j = K\mu_j - \sum_{i=0}^N a_i \binom{i}{j}.$$

The b_j are integers. Exactness and $0 \leq Kw_i - a_i < 1$ give

$$0 \leq b_j \leq \sum_{i=0}^N \binom{i}{j} = \binom{N+1}{j+1}.$$

The transposed Pascal matrix on nodes $0, \dots, r$ is unimodular. Its inverse gives integer corrections

$$d_i = \sum_{j=i}^r (-1)^{j-i} \binom{j}{i} b_j \quad (0 \leq i \leq r),$$

which satisfy $\sum_{i=0}^r d_i \binom{i}{j} = b_j$ for every $j \leq r$. Put $k_i = a_i + d_i$ for $i \leq r$, and $k_i = a_i$ otherwise. The assumed bound and $a_i \geq K\eta - 1$ imply $k_i \geq 0$. The corrected binomial moments are exact; these polynomials span the degree- r space. The zeroth moment gives $\sum_i k_i = K$. $\square$

Lemma 3 (The required moment denominators). *For integer N and $0 \leq j \leq r$, the denominator of μ_j divides*

$$r! \operatorname{lcm}(1, 2, \dots, r+1).$$

In particular $((r+1)!)^2 \mu_j$ is an integer.

Proof. $j! \binom{Nx}{j}$ is a polynomial in x with integer coefficients. Integrating its monomials introduces only denominators $1, \dots, j+1$. Thus the denominator divides $j! \operatorname{lcm}(1, \dots, j+1)$, which divides the displayed quantity and then $((r+1)!)^2$. $\square$

Lemma 4 (A uniform positive starting quadrature). *For every $r \geq 1$ and $N = (r+1)^2$, the Wilson–Hahn projection weights give an exact degree- r uniform quadrature on i/N satisfying*

$$w_i > \frac{1}{8N} \quad (0 \leq i \leq N).$$

Proof. Use the manuscript's Hahn polynomials with largest grid index N . Its estimates apply for $N \geq r(r+2)$, and show, for $r \geq 2$, that the unnormalized weights satisfy

$$W_i > \frac{N - 3r(r+3)/4}{N+1} > \frac{1}{8}.$$

The last inequality follows from $7(r+1)^2 - 6r(r+3) - 1 = (r-2)^2 + 2 > 0$. For $r = 1$, all $W_i = N/(N+1) > 1/8$ directly. Normalize by N . The applicability of the manuscript estimates uses $r(r+1) \leq N$ for Zaremba's nodal bound and $(r+1)^2 \leq N+1$ for Wilson's integral estimate. $\square$

Lemma 5 (An absolute scaling constant suffices). *For every integer $s \geq 2$,*

$$\frac{8s^2 \left(2^s \binom{s^2+1}{s} + 1 \right)}{(s!)^2} < 2^{25}.$$

Proof. The second summand, $8s^2/(s!)^2$, is at most 8. For the first, use

$$\binom{s^2+1}{s} \leq \frac{s^{2s} e^{1/s}}{s!}, \quad s! \geq \sqrt{2\pi s} (s/e)^s.$$

It is consequently at most

$$\frac{8e^{1/s}}{(2\pi)^{3/2}} \sqrt{s} \left(\frac{2e^3}{s} \right)^s < \sqrt{s} \left(\frac{81/2}{s} \right)^s.$$

Here $\pi > 3$ and $e < 2.72$ justify both strict inequalities, uniformly for $s \geq 2$. Since $\sqrt{s} < 2e^{s/16}$ and $\sup_{u>0} (a/u)^u = e^{a/e}$, this is bounded by

$$2 \exp \left(\frac{81}{2} e^{-15/16} \right) < 2e^{16}.$$

The inequality $e^{15/16} > 81/32$ follows by retaining its first five Taylor terms. Finally $2e^{16} + 8 < 2(11/4)^{16} + 8 < 2^{25}$. $\square$

Proof of Theorem 1. Start with the one-letter string $s_1 = 1^C$, where $C = 2^{25}$. Suppose s_r is r -fair and every letter occurs $C(r!)^2$ times. Set

$$N = (r+1)^2, \quad K = C((r+1)!)^2.$$

The old letters each occur $NC(r!)^2 = K$ times in N copies of s_r . The moment-denominator lemma says $K\mu_j$ is integral for every $j \leq r$. Use the starting quadrature from Lemma 4. Because $N = (r+1)^2$, the binomial coefficients $\binom{N+1}{j+1}$ increase for $0 \leq j \leq r$, and hence

$$\sum_{j=i}^r \binom{j}{i} \binom{N+1}{j+1} \leq \binom{N+1}{r+1} \binom{r+1}{i+1} \leq 2^{r+1} \binom{N+1}{r+1}.$$

Lemma 5, with $s = r+1$, verifies the sufficient positivity condition in Lemma 2, using $\eta = 1/(8N)$. Obtain integer gap lengths k_i of total K and exact degree- r Bernstein moments. The insertion criterion proves that the resulting string s_{r+1} is permutation-fair. Its new and old letters all occur K times, completing the induction. $\square$

Corollary 6 (Optimal leading exponent within whole-copy insertion). *The constructed dice satisfy*

$$\log F_n = 2n \log n - 2n + O(\log n).$$

The companion insertion-barrier note proves that every one-letter-at-a-time whole-copy insertion construction has an oldest die with $\log F_n \geq 2n \log n - O(n)$. Thus the leading coefficient 2 is optimal for that framework.

The constant is deliberately conservative. An exact finite check of the displayed sufficient ratio for $2 \leq s \leq 128$ gives a maximum below 2^{22} (near $s = 15$); an additional tail estimate would certify that smaller constant. This optimization has no bearing on the asymptotic result.